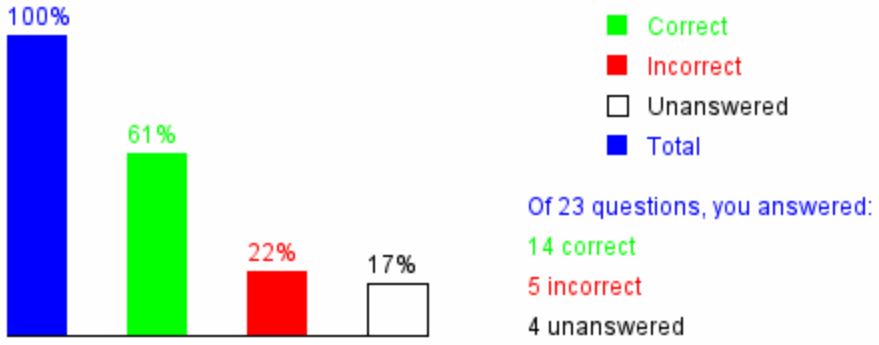
Introduction to Programming Using Python, Y. Daniel Liang

Many questions in this edition have been updated in the new edition. Please check with the publisher on the newest edition.

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Chapter 12 Inheritance and Polymorphism



Please send suggestions and errata to Dr. Liang at y.daniel.liang@gmail.com. Indicate which book and edition you are using. Thanks!

Section 12.2 Superclasses and Subclasses

12.1 Analyze the following code:

```
class A:
    def __init__(self, i = 0):
        self.i = i

class B(A):
    def __init__(self, j = 0):
        self.j = j

def main():
    b = B()
    print(b.i)
    print(b.j)

main()
```

- (●) A. Class B inherits A, but the data field in i in A is not inherited.
- ( ) B. Class B inherits A and automatically inherits all data fields in A.
- ( ) C. When you create an object B, you have to pass an integer such as B(5).
- ( ) D. The data field j cannot be accessed by object b.

Your answer is correct ✔

12.2 What will be displayed by the following code?

```
class A:
    def __init__(self, i = 1):
        self.i = i

class B(A):
    def __init__(self, j = 2):
        super().__init__()
        self.j = j

def main():
```

```
    b = B()
    print(b.i, b.j)

main()
```

- ○ A. 0 0
- ○ B. 0 1
- ◉ C. 1 2
- ○ D. 0 2
- ○ E. 2 1

Your answer is correct ✔

12.4 Suppose A is a subclass of B, to invoke the __init__ method in B from A, you write _____.

- ☑ A. `super().__init__()`
- ☐ B. `super().__init__(self)`
- ☐ C. `B.__init__()`
- ☑ D. `B.__init__(self)`

Your answer is correct ✔

12.7 Which of the following statements is true?

- ☑ A. A subclass is a subset of a superclass.
- ☑ B. When invoking a constructor from a subclass, its superclass's no-arg constructor is always invoked.
- ☑ C. You can override a non-private method defined in a superclass.
- ☑ D. You can override the initializer defined in a superclass.
- ☐ E. You can override a private method defined in a superclass.

Your answer ABCD is incorrect ✘
Hint: The constructor __init__ is a method. So, it can be overridden.
Click here to show the correct answer

Section 12.4 The object Class

12.8 What will be displayed by the following code?

```
class A:
    def __new__(self):
        self.__init__(self)
        print("A's __new__() invoked")

    def __init__(self):
        print("A's __init__() invoked")


class B(A):
    def __new__(self):
        print("B's __new__() invoked")

    def __init__(self):
        print("B's __init__() invoked")


def main():
    b = B()
    a = A()
```

```
main()
```

- ○ A. `B's __new__() invoked and followed by A's __init__() invoked`
- ○ B. `B's __new__() invoked followed by A's __new__() invoked`
- ◉ C. `B's __new__() invoked, followed by A's __init__() invoked, and followed by A's __new__() invoked`
- ○ D. `A's __init__() invoked and followed by A's __new__() invoked`

Your answer is correct 

12.9 Which of the following statements is true?

- ☑ A. `By default, the __new__() method invokes the __init__ method.`
- ☑ B. `The __new__() method is defined in the object class.`
- ☑ C. `The __init__() method is defined in the object class.`
- ☑ D. `The __str__() method is defined in the object class.`
- ☑ E. `The __eq__(other) method is defined in the object class.`

Your answer is correct 

Section 12.5 Polymorphism and Dynamic Binding

12.10 What will be displayed by the following code?

```
class A:
    def __init__(self):
        self.i = 1

    def m(self):
        self.i = 10

class B(A):
    def m(self):
        self.i += 1
        return self.i


def main():
    b = B()
    print(b.m())

main()
```

- ○ A. `1`
- ◉ B. `2`
- ○ C. `10`
- ○ D. `i is not accessible from b.`

Your answer is correct 

12.11 What will be displayed by the following code?

```
class A:
    def __str__(self):
        return "A"

class B(A):
    def __str__(self):
        return "B"
```

```
class C(B):
    def __str__(self):
        return "C"

def main():
    b = B()
    a = A()
    c = C()
    print(a, b, c)

main()
```

- ○ A. C C C
- ○ B. A B C
- ◉ C. A A A
- ○ D. B B B

Your answer C is incorrect 
Click here to show the correct answer

12.12 What will be displayed by the following code?

```
class A:
    def __str__(self):
        return "A"

class B(A):
    def __init__(self):
        super().__init__()

class C(B):
    def __init__(self):
        super().__init__()

def main():
    b = B()
    a = A()
    c = C()
    print(a, b, c)

main()
```

- ○ A. C C C
- ○ B. A B C
- ◉ C. A A A
- ○ D. B B B

Your answer is correct 

12.14 What will be displayed by the following code?

```
class Person:
    def getInfo(self):
        return "Person's getInfo is called"

    def printPerson(self):
        print(self.getInfo(), end = ' ')

class Student(Person):
    def getInfo(self):
```

```
        return "Student's getInfo is called"

def main():
    Person().printPerson()
    Student().printPerson()

main()
```

- ○ A. Person's getInfo is called Person's getInfo is called
- ◉ B. Person's getInfo is called Student's getInfo is called
- ○ C. Student's getInfo is called Person's getInfo is called
- ○ D. Student's getInfo is called Student's getInfo is called

Your answer is correct 

12.15 What will be displayed by the following code?

```
class Person:
    def __getInfo(self):
        return "Person's getInfo is called"

    def printPerson(self):
        print(self.__getInfo(), end = ' ')

class Student(Person):
    def __getInfo(self):
        return "Student's getInfo is called"

def main():
    Person().printPerson()
    Student().printPerson()

main()
```

- ◉ A. Person's getInfo is called Person's getInfo is called
- ○ B. Person's getInfo is called Student's getInfo is called
- ○ C. Student's getInfo is called Person's getInfo is called
- ○ D. Student's getInfo is called Student's getInfo is called

Your answer is correct 

12.16 Analyze the following code:

```
class A:
    def __init__(self):
        self.setI(20)
        print("i from A is", self.i)

    def setI(self, i):
        self.i = 2 * i;

class B(A):
    def __init__(self):
        super().__init__()

    def setI(self, i):
        self.i = 3 * i;


b = B()
```

- ○ A. The __init__ method of class A is not called.

B. The __init__ method of class A is called and it displays "i from A is 0".
C. The __init__ method of class A is called and it displays "i from A is 40".
D. The __init__ method of class A is called and it displays "i from A is 60".

Your answer is correct

12.17 Analyze the following code:

```
class A:
    def __init__(self):
        self.setI(20)

    def setI(self, i):
        self.i = 2 * i;

class B(A):
    def __init__(self):
        super().__init__()
        print("i from B is", self.i)

    def setI(self, i):
        self.i = 3 * i;


b = B()
```

A. The __init__ method of class A is not called.
B. The __init__ method of class A is called and it displays "i from B is 0".
C. The __init__ method of class A is called and it displays "i from B is 40".
D. The __init__ method of class A is called and it displays "i from B is 60".

Your answer C is incorrect
Click here to show the correct answer

Section 12.6 The isinstance Function

12.18 To check whether an object o is an instance of class A, use _____.

A. `o.isinstance(A)`
B. `A.isinstance(o)`
C. `isinstance(o, A)`
D. `isinstance(A, o)`

Your answer is correct

Section 12.8 Class Relationships

12.19 What relationship is appropriate for Company and Employee?

A. `association`
B. `composition`
C. `inheritance`

Your answer C is incorrect
Click here to show the correct answer

12.20 What relationship is appropriate for Course and Faculty?

- (●) A. `association`
- ( ) B. `composition`
- ( ) C. `inheritance`

Your answer is correct 

12.21 What relationship is appropriate for Student and Person?

- (●) A. `association`
- ( ) B. `composition`
- ( ) C. `inheritance`

Your answer A is incorrect 
Click here to show the correct answer

12.22 What relationship is appropriate for House and Window?

- ( ) A. `association`
- (●) B. `composition`
- ( ) C. `inheritance`

Your answer is correct 

12.23 What relationship is appropriate for Account and Savings Account?

- ( ) A. `association`
- ( ) B. `composition`
- (●) C. `inheritance`

Your answer is correct 